



2. Review of the nullspace and an introduction to the column space, a geometric example

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Let $\mathbf{f}$ be the function from $\mathbb{R}^3$ to $\mathbb{R}^3$ that projects all of $\mathbb{R}^3$ onto the xy -plane:

$$\mathbf{f} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

1. What is the matrix $\mathbf{A}$ that represents $\mathbf{f}$?
2. Which vectors from the input vector space does $\mathbf{f}$ map to $\mathbf{0}$?
3. Describe the range (or image) of $\mathbf{f}$ geometrically in the output space. Recall that the range of $\mathbf{f}$ is the set of all vectors $\mathbf{b} = \mathbf{f}(\mathbf{x})$ for all vectors $\mathbf{x}$ in the input space.

Solution:

1. $\mathbf{A}$ is a 3×3 matrix such that

$$(\text{first column of } \mathbf{A}) = \mathbf{f} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$(\text{second column of } \mathbf{A}) = \mathbf{f} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

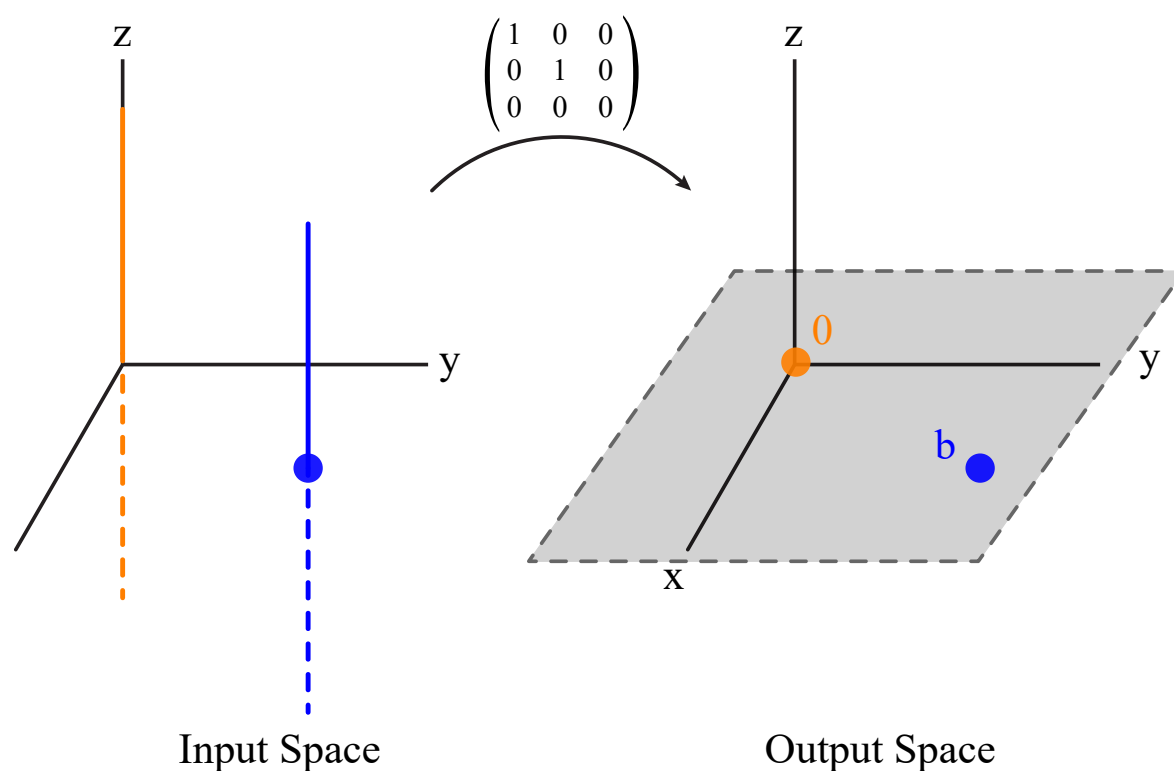
$$(\text{third column of } \mathbf{A}) = \mathbf{f} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

$$\text{Thus } \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

2. Note that $\mathbf{f} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$ is the zero vector if and only if $x = 0$ and $y = 0$. Thus the

projection takes any vector of the form $\begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}$ to the zero vector. Geometrically, this says that

the entire z -axis is sent to the zero vector.



The set of vectors mapped to $\mathbf{0}$ by $\mathbf{f}$ is the same as the set of solutions to $\mathbf{Ax} = \mathbf{0}$. Using our algorithm from the last lecture, we find the set of solutions to $\mathbf{Ax} = \mathbf{0}$. This set of solutions forms a vector space called the nullspace. The nullspace $\mathbf{NS}(\mathbf{A})$ is a subspace of the **input space**; in this example:

$$\mathbf{NS}(\mathbf{A}) = \{\text{solutions to } \mathbf{Ax} = \mathbf{0}\}$$
$$= \text{solutions to } \mathbf{f}(x, y, z) = \mathbf{0}$$
$$= \{(0, 0, z) : z \in \mathbb{R}\}$$
$$= \text{the } z\text{-axis in the input space } \mathbb{R}^3.$$

3. The set of all vectors $\mathbf{b}$ in the image (also called the range) of $\mathbf{f}$ are all vectors $\mathbf{f}(\mathbf{x})$; in our example:

$$\text{Image of } \mathbf{f}(x, y, z) = \{(x, y, 0) : x, y \in \mathbb{R}\}$$
$$= \text{the } xy\text{-plane in the output space } \mathbb{R}^3.$$

Another way of thinking of this is as the set of all vectors $\mathbf{b}$ that can be written as $\mathbf{Ax}$ for some vector $\mathbf{x}$; in our example:

$$\{\text{All vectors } \mathbf{Ax}\} = \text{all linear combinations of the columns of } \mathbf{A}.$$

Therefore the set of all vectors $\mathbf{b} = \mathbf{Ax}$ is the same as the span of the columns of $\mathbf{A}$. The span of the columns of $\mathbf{A}$ is called the **column space $\mathbf{CS}(\mathbf{A})$** ; it is a subspace of the **output space**.

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